

Data Warehouse Design: “Mexico City Urban Intelligence”

Business Intelligence — Project 2

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I. INTRODUCTION

Information about a city does not usually come from one single place. Population, business activity and crime data are collected and published separately, which makes it hard to see how they relate to each other in space. This project integrates three public sources for Mexico City into a PostgreSQL/PostGIS dimensional warehouse, and then uses it to answer two questions: how are population, business activity and crime related across the city, and are those patterns spatially clustered? To answer them, the work was divided into three phases: a data and geographic assessment, a reproducible ETL with a star-schema warehouse, and a spatial analysis that includes correlations, global Moran's I, LISA and a bivariate Moran statistic.

The study area is Mexico City (CDMX), where the prosecutor's office (Fiscalía General de Justicia) publishes the coordinates of every reported crime [4]. All the code, SQL, notebooks and outputs are stored in one Git repository, so the whole work can be reproduced from the raw downloads by following the short list of commands in the README.

The rest of this report is organized as follows. Section II states the objectives, Section III describes the data sources and the geographic strategy, Section IV presents the ETL and the warehouse architecture, and Section V explains the spatial methods. Section VI reports the results, Section VII lists the limitations, and Section VIII concludes.

Before running the whole analysis that was made carefully, some assumptions were made about what the data would show and how maybe would interact, for example that crime

would be highest where most people live logically. These expectations were not tested formally, but they are surely compared with the results in Section VI as it was said before, because several of them turned out to be different from what was first expected as in any kind of study. First we have an hypothesis and then it has to be prove it where it can be the result positive or negative.

II. OBJECTIVES

A. General Objective

To design and implement a reproducible geospatial data warehouse that integrates demographic, economic and public-safety data for Mexico City, and to analyze how these dimensions are distributed and related in space.

B. Specific Objectives

- 1) **Data and geographic assessment:** Profile every source and choose a geographic unit at which all the sources can be joined.
- 2) **ETL:** Clean the sources and load them into PostGIS through reproducible steps with automatic validation checks.
- 3) **Dimensional model:** Build a star schema with its facts, dimensions and grain clearly documented.
- 4) **KPIs:** Compute the 14 required indicators directly from the warehouse using SQL.
- 5) **Spatial analysis:** Test at least three correlations, calculate global Moran's I for at least two indicators, find LISA clusters and apply a bivariate spatial statistic.

III. DATA SOURCES AND GEOGRAPHIC STRATEGY

A. Data Sources

Table I lists the four sources used in this project. These sources do not share the same year. The Census is decennial and the 2020 edition is the only one available, so population comes from 2020. Because of this, DENUÉ 11/2024 and the 2024 crime data were chosen so that, at least, businesses and crime describe the same year.

TABLE I: Data sources

Source	Content and grain	Year
INEGI Census [1]	Population and economic indicators; 2,433 AGEb total rows	2020
INEGI DENUÉ [2]	458,228 establishments (points, SCIAN activity, size)	11/2024
FGJ CDMX [4]	216,965 files with incident date in 2024 (points, date, hour, offense)	2024
INEGI Marco Geoestadístico [3]	2,431 urban AGEb polygons and state boundary	2020

B. Choosing the Geographic Unit

Each source describes geography in a different way: the Census gives areas, DENUÉ gives points plus area codes, and the crime file gives points only. Because of this, four candidate units were compared before choosing one. The 16 alcaldías are too few observations to run spatial statistics. City blocks (66,456) are too fine, since most of them have no crime and the Census hides many of their values. Colonias were not an option because the sources do not include official polygons for them. The **urban AGEb** was the best compromise: INEGI publishes official polygons for it, the Census reports a complete set of indicators at this level, and with 2,431 areas each AGEb averages about 3,759 residents and 81 crime files.

C. Integration Rules and Test

The Census is joined to the polygons through the 13-character key CVEGEO, which is built from the state, alcaldía, locality and AGEb codes. Crime and DENUÉ points, on the other hand, are assigned to polygons with a spatial join (`ST_Intersects` in PostGIS); in the case of DENUÉ, the AGEb code included in the file is only used as a fallback. All data are stored in EPSG:4326, and areas are computed in EPSG:32614 (UTM zone 14N). In the Phase 1 test, 99.7% of the georeferenced crime points and 99.7% of the establishments fell inside exactly one AGEb, no point matched two polygons, and the DENUÉ AGEb code agreed with the spatial join for 99.8% of the establishments. The few points that were not matched are located in rural land outside the urban layer. Finally, two Census AGEb (7,108 residents) have no polygon, so they were excluded.

IV. ETL AND WAREHOUSE ARCHITECTURE

A. ETL Pipeline

The ETL is made of a set of numbered Python scripts followed by SQL scripts. The raw files are never modified: each source is cleaned and saved as a parquet or GeoPackage

file, written to a staging schema, and then loaded into the schema `dw` with SQL. The load is idempotent, and a validation script compares the warehouse with the staged data (row counts, population totals, foreign keys, SRIDs and documented columns). If any of its 11 checks fails, the pipeline stops. The main decisions made during this process were:

- Suppressed Census values (* and N/D) are converted to NULL and never to zero. The 25–59 age group is not published, so it was derived as the total minus the other groups.
- Crime is filtered by the date of the incident and not by the filing year, because the two can be different. In total, 6,504 non-criminal events were removed, and 12,403 files without coordinates were kept in staging, although they are not mapped.
- The FGJ category is too coarse, so 10 analytical crime groups were built from the offense name.
- DENUÉ establishments with coordinates outside CDMX are flagged as invalid and, when possible, assigned to an AGEb through their AGEb code.

B. Star Schema

The model is a star schema centered on the AGEb dimension (Fig. 1). Each fact table keeps its native grain: one row per AGEb for population, one row per AGEb and age group for age, one row per establishment for DENUÉ, and one row per incident for crime. Keeping the lowest grain means that any KPI can be rebuilt at another level without having to reload the data. In addition, `dim_ageb` stores the polygon in two coordinate systems, one for display and another for metric calculations, and all the geometries have GiST indexes.

C. KPIs

All 14 required KPIs are computed in SQL, using the view `dw.vw_ageb_kpis` and a second view for crime by type and time (Table II). Rates are only analyzed for AGEb with at least 500 residents, since rates calculated from very small populations are unstable.

TABLE II: Required KPIs and formulas

#	KPI	Formula
1	Total population	Census total (POBTOT)
2	Population density	population / area (km ²)
3	Economic activity rate	100 × PEA / population 12+
4	Population by age group	residents in 8 age groups
5	Total businesses	establishments in the AGEb
6	Business density	businesses / km ²
7	Businesses per 1,000	1,000 × businesses / population
8	Retail density	retail (SCIAN 46) per km ²
9	Service density	services (SCIAN 51–56, 61, 62, 71, 72, 81) per km ²
10	Dominant activity	SCIAN sector with most establishments
11	Total crime incidents	incidents with date in 2024
12	Crime rate	1,000 × incidents / population
13	Incidents by type and time	crime group, month, weekday, time band
14	Crime vs. business activity	100 × incidents / businesses

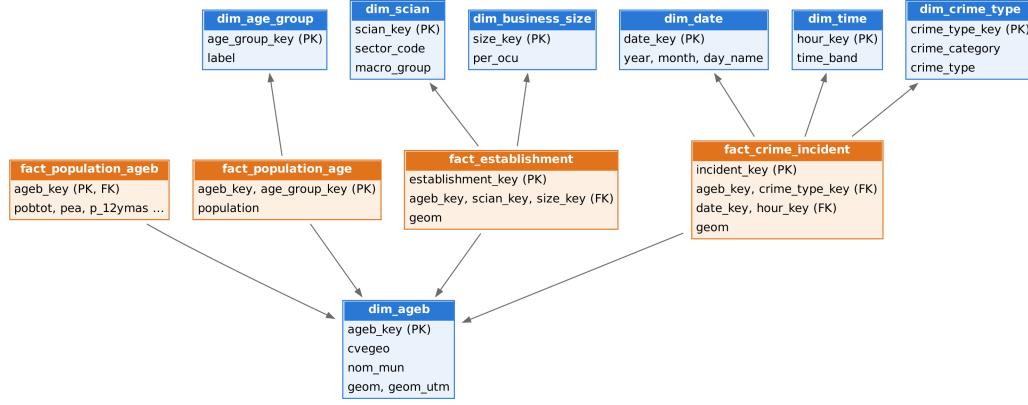


Fig. 1: Star schema of the warehouse. Orange tables are facts and blue tables are dimensions. Four facts share the AGEB dimension.

V. SPATIAL ANALYSIS METHODS

The spatial statistics use a Queen contiguity matrix (row-standardized) as the main neighbor rule, and the 8 nearest neighbors (KNN-8) as a sensitivity check, with 999 permutations for inference. Because the variables are skewed, Spearman correlation was used as the main measure, and Pearson correlation on log-transformed values is reported as a complement. Global Moran's I measures the overall clustering, while LISA (local Moran's I) finds local clusters, labeled High-High, Low-Low, High-Low and Low-High. The bivariate Moran statistic relates the business density of an AGEB with the crime rate of its neighbors. Finally, as a robustness check, the Moran test was repeated with Empirical Bayes smoothed rates, which reduce the influence of small populations.

VI. RESULTS

A. Geographic Distribution

In total, the warehouse holds 9,138,524 residents, 456,763 establishments and 197,518 located crime incidents. Population density is high across much of the north and east of the city, while business density is concentrated in the central alcaldías (Fig. 2). Retail trade is the dominant activity in 87% of the AGEB. In terms of crime, Cuauhtémoc (28,795 incidents) and Iztapalapa (27,212) have the most incidents, but the ranking changes once population is taken into account: Cuauhtémoc has 52.7 incidents per 1,000 residents, while Iztapalapa, the most populated alcaldía, has 14.8. Robbery is the largest crime group (67,741 files, 32% of the total), followed by family violence and fraud and extortion.

B. Correlations

Table III shows five pairs of indicators calculated with 2,311 AGEB. Business density and population density are moderately correlated. The crime rate is positively related to business density, more weakly related to the economic activity rate and to retail density, and most strongly related to the share of residents aged 60 or more. All of these relations are statistically significant, but every coefficient is below

0.4, which means they describe tendencies and not strong relations, and none of them says anything about cause. The age result probably reflects where older residents live and not an effect of age on crime, although this explanation was not tested.

TABLE III: Correlations between AGEB indicators ($n = 2,311$)

Pair	Spearman ρ	Pearson r (log)
Business density – population density	0.39	0.44
Crime rate – business density	0.28	0.32
Crime rate – economic activity rate	0.18	0.27
Crime rate – share aged 60+	0.40	0.33
Crime rate – retail density	0.07	0.15

C. Spatial Autocorrelation

Global Moran's I is positive and significant for every indicator (pseudo $p = 0.001$, Table IV). The KNN-8 weights give the same picture, so the result does not depend on how the neighbors are defined. This means that nearby AGEB are much more similar to each other than a random arrangement would produce, which is why ordinary statistics that assume independent observations should be read with care. Fig. 3 shows the Spearman correlation matrix and the Moran scatter plot of the crime rate, whose slope is equal to Moran's I.

TABLE IV: Global Moran's I (999 permutations, all $p = 0.001$)

Indicator	Queen	KNN-8
Economic activity rate	0.53	0.50
Population density (log)	0.49	0.47
Crime rate per 1,000 (log)	0.49	0.45
Business density (log)	0.43	0.37

D. Hot Spots and the Role of Business Activity

LISA (local Moran's I) was used to find where the clusters of crime are located, instead of only knowing that crime is clustered in general. It finds 248 High-High AGEB, which are areas with high crime that are surrounded by other areas with high crime, and 191 Low-Low AGEB, which are areas

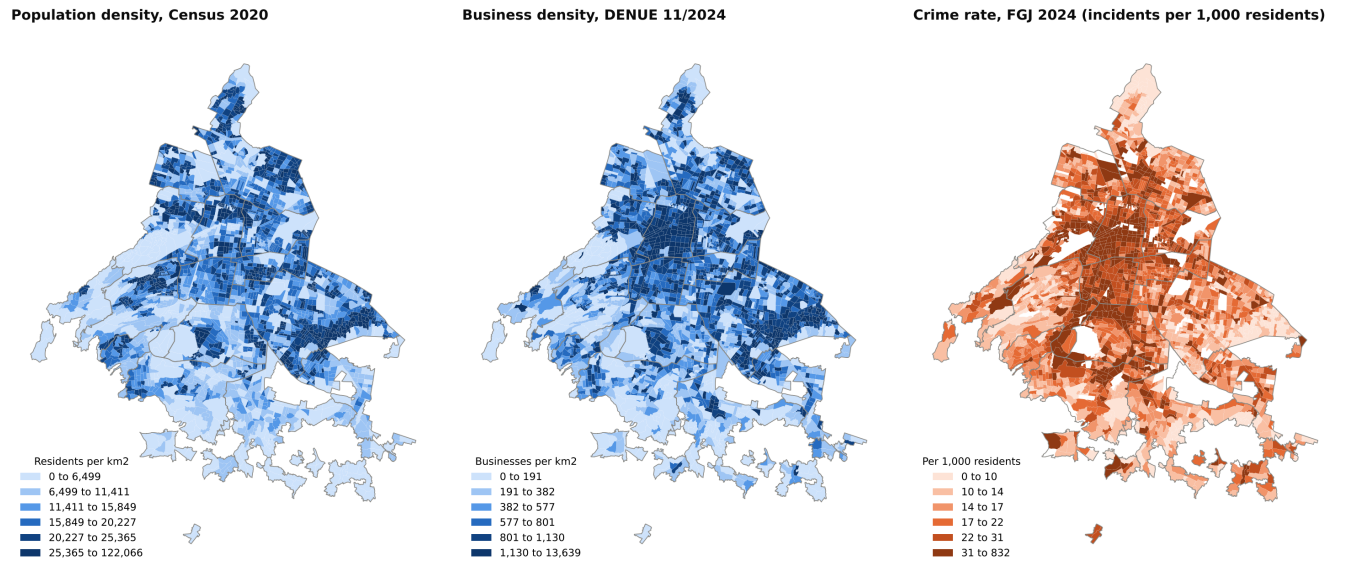


Fig. 2: Population density (Census 2020), business density (DENUE 11/2024) and crime rate (FGJ 2024) by AGEB, in quantile classes.

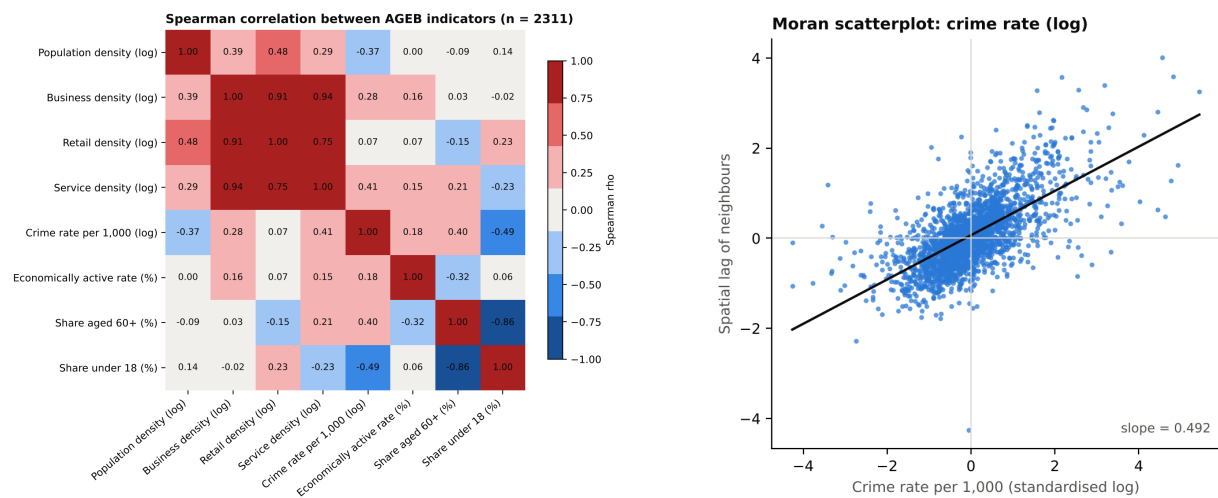


Fig. 3: Spearman correlation matrix (left) and Moran scatter plot of the crime rate, slope 0.49 (right).

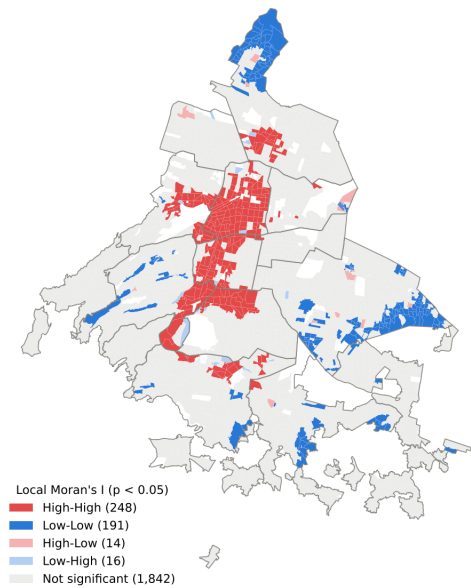
with low crime surrounded by other areas with low crime (Fig. 4, left). The High-High areas are the ones that can be considered hot spots.

These hot spots are not spread randomly across the city. They form one connected region in the center, and most of them are concentrated in only a few alcaldías. Cuauhtémoc alone has 95 of them, which is 67% of its AGEB, and Cuauhtémoc, Benito Juárez and Miguel Hidalgo together hold 166 of the 248. In other words, the high crime of the city is not found in many separate places, but mostly in one central zone.

The report also asks whether business activity has something to do with this pattern. For this, the bivariate Moran’s I was used, which compares the business density of an AGEB with the crime rate of its neighbors. Its value is 0.19 ($p = 0.001$). This means that AGEB with many businesses tend

to be surrounded by neighbors with higher crime. However, the relation is much weaker than the crime pattern itself, so business density helps to describe the picture but does not explain it by itself. The local version of this statistic (Fig. 4, right) shows that this association is concentrated in the same central region where the hot spots are. Finally, a check was made to be sure that these clusters were real. Areas with few residents can have very unstable crime rates, and this could create clusters that do not actually exist. To test this, the analysis was repeated with Empirical Bayes smoothed rates, which reduce the influence of small populations. The autocorrelation stays clearly positive ($I = 0.37$), so the clustering is not an artifact of unstable rates.

LISA clusters: crime rate per 1,000 residents (2024)



Local bivariate Moran: business density vs crime rate in neighbours

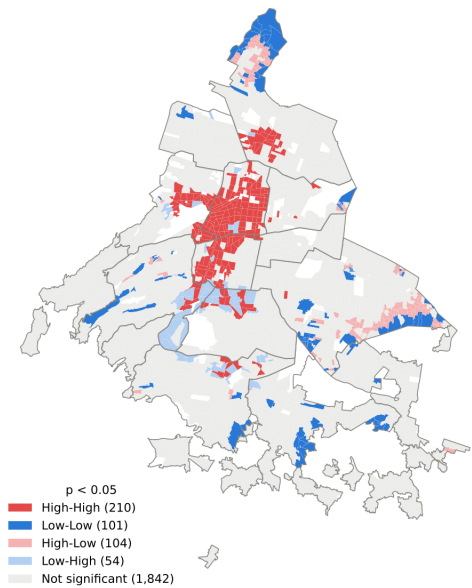


Fig. 4: Left: LISA clusters of the crime rate. Right: local bivariate Moran, business density against the crime rate of neighboring AGEB.

E. What We Expected, What We Found and What It Means

Before running the analysis, the answers were not known, and several results turned out to be different from what was first expected. In any project that works with real data, it is common to start with some ideas about what the results will look like, even if those ideas are never written down as formal hypotheses.

In this case, those ideas were ordinary assumptions about the quality of the data, about where crime would be highest, and about the role of businesses and older residents. Table V puts the main expectations next to what the data showed

and what each difference means for this project. These expectations are not hypotheses that were tested formally, but they are included because comparing them with the results helps to understand what was learned from the analysis.

Due to all that is the differences are the most useful part of the analysis done. If every result had matched the initial expectations, the warehouse would not have shown anything new. The surprises (the rate in the center, the weak role of businesses and the age result) also show the limits of working with area totals, which is why they are described as associations in the next section.

TABLE V: Expectations compared with the results

What we expected	What we found	What it means for us
The data would be ready to use once downloaded.	About 6% of the 2024 criminal files have no coordinates, 44% of the crime times fall on round hours and the Census hides some values.	A large part of the work was cleaning and checking the data, not modeling. Every fix is documented in the ETL, and the validation checks make any failed load visible.
Crime would be highest where most people live, so Iztapalapa should lead.	Cauhtémoc has the most incidents in total and, by far, the highest rate (52.7 per 1,000 residents against 14.8 in Iztapalapa).	Counts and rates tell different stories. The rate in the center is probably inflated by workers and visitors who are not residents, so it should not be read as “more dangerous” without taking that into account.
More businesses would clearly mean more crime.	Only a weak to moderate link: a Spearman coefficient of 0.28 with business density, 0.07 with retail density, and 0.19 for the bivariate Moran statistic.	Business activity is part of the picture, but it is not a strong predictor. The type of business and the hour of the day were not analyzed, and they may matter more.
The share of older residents would have little to do with crime.	It was the strongest correlation of the five (0.40).	This is read as a geographic coincidence, because older residents are not spread evenly across the city, and not as an effect of age. This explanation was not tested.
Hot spots would appear in several places, including the most populated alcaldías.	One connected region in the center: Cauhtémoc has 95 of the 248 High-High AGEB, and three central alcaldías hold 166 of them.	Crime is organized in space, so treating each AGEB as independent would be misleading. The map shows where to look, but it does not explain why.
Noisy rates in small AGEB might be creating the clusters.	Moran’s I fell from 0.49 to 0.37 with Empirical Bayes smoothing, but it stayed clearly positive.	Part of the raw clustering came from unstable rates, but most of it is real.

VII. LIMITATIONS

Some limitations that were found during the project were the following:

- **Different years.** Population comes from 2020, while businesses and crime come from 2024, so the rates mix different periods and do not take population change into account.
- **Reports, not crimes.** The FGJ file contains reported cases, not all the crimes that happened. Reporting varies across areas, and about 6% of the 2024 criminal files have no coordinates.
- **Heaped times.** 44% of the crime times fall exactly on a round hour, so time bands were used instead of single hours.
- **Resident denominator.** Crime per resident is probably inflated in central AGEB that have many workers and visitors but few residents. For this reason, crime per business is also reported, although the floating population was not modeled.
- **Modifiable areal unit.** AGEB differ in size and shape, so the results could change if another unit were used. Two neighbor rules and smoothed rates were tested, but not another geographic unit.
- **Association only.** Correlations and Moran statistics describe patterns, so they cannot show that businesses cause crime or the other way around.

VIII. CONCLUSION

This project built a reproducible geospatial data warehouse for Mexico City by integrating three public sources (the Census 2020, DENUe and the FGJ crime data) into a PostgreSQL/PostGIS star schema. The urban AGEB was chosen as the common geographic unit, and the integration was verified before running the analysis: 99.7% of the crime points and establishments were assigned to exactly one AGEB. The warehouse computes all 14 required KPIs with SQL, and its validation checks make any failed load visible, so the whole process can be reproduced from the raw downloads.

The spatial analysis showed that population, business activity and crime are all spatially clustered, which means that nearby AGEB are much more similar to each other than a random arrangement would produce. In the case of crime, this pattern is concentrated in one large hot spot in the central alcaldías, where Cuauhtémoc alone has 95 of the 248 High-High AGEB. The clustering is not an artifact of unstable rates, since Moran's I stayed clearly positive (0.37) after applying Empirical Bayes smoothing. Business density, on the other hand, was only modestly associated with crime in neighboring areas (bivariate Moran's $I = 0.19$), so it is part of the picture but not a strong explanation by itself.

Several results were different from what was first expected (Section VI-E). The data was not ready to use once downloaded, and a large part of the work was cleaning and checking it instead of modeling. The most populated alcaldía, Iztapalapa, is not the one with the highest crime rate (14.8

incidents per 1,000 residents against 52.7 in Cuauhtémoc). Also, the share of older residents had the strongest correlation with crime, which is read as a geographic coincidence and not as an effect of age. These differences are the most useful part of the project, because they show the limits of working with area totals.

The main limits of this work are the gap between the Census year and the other sources, the use of residents as the denominator, and the fact that only associations are described, so none of these results shows that one factor causes another. Future work includes adding a time dimension to the warehouse (monthly crime and a daytime population estimate) and repeating the analysis with a second geographic unit, to check whether the patterns remain the same.

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